**DO NOW — RECALL**

# Algorithms: synoptic exam challenge

Name: _____ Class: _____ Date: _____

**1. True remaining cost is 4; is h(A)=7 admissible?**

**2. If g=3 and h=4, what is f?**

**3. What condition is needed for Dijkstra's standard shortest-path guarantee?**

**4. When partition sets the pivot at index 2, which recursive index ranges remain within indexes 0–4?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

No. 7 is an overestimate.

### 2. Model answer

7.

### 3. Model answer

Non-negative edge weights.

### 4. Model answer

0–1 and 3–4.

### Vocabulary — know these terms

partition — Rearrange around a pivot and place that pivot.

tentative distance — Current best-known cost from the start.

predecessor — Vertex immediately before another on a path.

heuristic — Estimate of remaining distance to the goal.

admissibility — A heuristic never exceeds the true cheapest remaining cost.

NEW LEARNING

# Read, annotate and explain

### Match the algorithm to the task

Quicksort partitions an array. Dijkstra finds shortest routes on non-negative weighted graphs. A* adds a remaining-cost estimate.

### Show all required intermediate states

For quicksort write arrays after partition. For shortest paths maintain distances, predecessors and a clear visited or frontier order.

### Explain efficiency and preconditions

State when each algorithm is appropriate and how its method changes work.

### Exam command words matter

Trace = show states. Explain = give a cause or reason. Compare = connect similarities and differences with evidence.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Use last-element Lomuto partition on [5,2,4]. Give final pivot index.

### Predict first

Start with pivot 4; 5 is too large to move to the smaller section.

### Model response

Move 2 to the left: [2,5,4].

### Model response

Swap pivot into index 1: [2,4,5]. Only one partition is being demonstrated.

**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Using last-element Lomuto quicksort on [7,2,6,1,5]: (a) state initial pivot [1]; (b) show the array after moving each qualifying value [3]; (c) give array and pivot index after partition [2]; (d) give fully sorted array [2]; (e) state typical average and worst-case time complexity [2].

### Question 2

Find the shortest route A→E for non-negative edges: A–B=2, A–C=6, B–C=1, B–D=7, C–D=2, D–E=3, C–E=9. Give tentative-distance improvements [8] and final path [2].

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Graph edges S–A=2, S–B=5, A–B=1, B–G=3, A–G=7. Heuristics h(S)=5,h(A)=3,h(B)=2,h(G)=0. (a) Define g, h, f [3]; (b) show initial f for A and B [2]; (c) update B after expanding A [2]; (d) state shortest route and cost [2]; (e) evaluate h(A) admissibility [1].

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Quicksort, Dijkstra and A* are interchangeable because they all have a loop.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.